CryWatch

On-device AI that understands why your baby is crying

ASUS UGen AI League 2026 — Everyday AI · Battlefield Lightning · Global Online Track

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The 3 AM question

Every parent knows this sound. Nobody knows what it means.

- Is she hungry? Tired? In pain? Too cold? Guess wrong and nobody sleeps.
- Not understanding cries is the most-cited stressor for first-time parents.
- Today's “smart” monitors stream nursery audio to someone else's cloud servers —
- your baby's most vulnerable moments, stored who-knows-where, behind a subscription.

Parents need answers, not audio streams. And the audio should never leave the room.

Why now

Edge AI finally fits the nursery.

- The ASUS UGen300 puts 40 TOPS (INT4) of neural compute on a 2.5 W USB stick —
- enough to run audio AI continuously, on-device, for pennies of electricity.
- Parents are rejecting cloud baby-tech over privacy fears — on-device is the differentiator.
- Cry-reason classification is an unsolved consumer problem: detection exists, understanding doesn't.

CryWatch

Hear the cry. Know the reason. Keep it private.

- A nursery microphone listens locally — nothing is recorded to the cloud, ever.
- A tiny neural net classifies each cry into 9 reasons: hungry, tired, belly pain, discomfort,
- burping, too cold/hot — and tells cry apart from laughter, background noise, and silence.
- Each prediction ships with parent-friendly guidance: “likely hunger — check when the last feed was.”
- Runs in milliseconds on CPU today; Stage II target: the full pipeline on the UGen300.

It works — measured, not promised

Held-out test set. Clips the model never saw in training.

TEST ACCURACY

44.0%

9 cry reasons + background

CRY DETECTION

98.4%

cry vs laugh / noise / silence

MODEL SIZE

~51K params

<300 KB fp32 · ~70 KB int8

Training: 1,305 labeled clips · stratified 70/15/15 split · class-weighted loss · full code on GitHub.

Per-class test accuracy: belly pain 28% · burping 44% · cold_hot 13% · discomfort 5% · hungry 33% · laugh 100% · noise 75% · silence 100% · tired 37%

How it works

Classical signal processing + a tiny learned classifier.

The front-end is fixed math (no learned weights); only the 70K-param MLP is learned — which is exactly the part that quantizes cleanly to INT8 for the Hailo-10H NPU.

The model

Small on purpose: built for the edge, not the datacenter.

- Architecture: 480 → 96 → 48 → 9, ReLU, softmax. 51,273 parameters.
- Footprint: ~280 KB fp32, ~70 KB after INT8 quantization — fits in NPU SRAM.
- Training: pure NumPy + SciPy, CPU-only, minutes — no GPU cluster, no API bills.
- Inference: milliseconds per 7-second clip on an ordinary laptop CPU.
- Imbalance handled: inverse-frequency class weights (hungry dominates the dataset).
- Reproducible: fixed seeds, stratified splits, every number regenerable from this repo.

Hardware / software architecture

Stage II target: the entire loop on the UGen300.

No cloud in the loop. Audio never leaves the home.

Interfaces: USB-C (UGen300) · Linux/Windows/Android hosts · PyTorch → ONNX → Hailo SDK

Expected outcomes

What winning looks like.

- Stage I: shortlist — working prototype, honest metrics, credible edge deployment plan.
- Stage II: INT8 model running live on the UGen300; nursery app demo; ≥90% cry-detection accuracy.
- Product: a $99-class nursery device or USB add-on for existing monitors — no subscription.
- Impact: fewer 3 AM guessing games; nursery audio stays in the nursery.
- Commercial path: license to baby-monitor makers; direct device via the UGen300 USB form factor.

Stage II roadmap

Nov 6 – Dec 4 development sprint.

- **Wk 1–2** Quantize to INT8, export ONNX, compile with the Hailo SDK; validate numerics vs fp32.
- **Wk 2–3** Deploy on the shipped UGen300; measure latency + power; tune the voice-activity gate.
- **Wk 3–4** Build the nursery companion app: alerts, reason + guidance, cry diary.
- **Wk 4** 10-minute final demo video, updated deck, full repo — final submission Dec 4.

Team

One builder, personal stake, AI-assisted engineering.

- Dakota McCall — builder. Dad of a 5-month-old; this product exists because his nursery needs it.
- Engineering: AI-assisted development (Muse) — every line reviewed, every metric reproduced from the repo.
- Track: Global Online — judged remotely, no travel required.

References

- ASUS UGen300 M2/USB AI Module — Hailo-10H, 40 TOPS INT4, 2.5 W (asus.com).
- ASUS UGen AI League Innovation Hackathon — official rules (contest platform).
- Hailo-10H overview — Hailo developer zone; HailoRT SDK + ONNX support.
- Baby cry sound dataset — 1,197 labeled clips, 9 classes (Hugging Face, mahmudulhasan01).
- Infant cry acoustics — fundamentals 300–600 Hz; reason-from-cry classification literature.

Code & docs: github.com/midnightmichigan/crywatch